

ESC/POS Command Set

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Explain: this command centralized dot unit = 0.125 mm (1mm=8dots) ; value “n” in decimal

1 printer command

1.1 ESC @

[Name] Initialize printer

[Format]	ASCII	ESC	@
	Hex	1B	40
	Decimal	27	64

[Description] initialize printer and recover printer to default parameter

1.2 GS FF

[Name] Feed paper to print starting position (only used for black mark paper)

[Format]	ASCII	GS	FF
	Hex	1D	0C
	Decimal	29	12

[Description] print out all data in buffer, if blackmark is valid , then printer will feed paper to blackmark position.

1.3 LF

[Name] Print and line feed

[Format]	ASCII	LF
	Hex	0A
	Decimal	10

[Description] Prints data in buffer and feeds one line based on the current line spacing

1.4 ESC J n

[Name] Print and feed paper

[Format]	ASCII	ESC J	n
	Hex	1B 4A	n
	Decimal	27 74	n

[Range] $0 \leq n \leq 255$

[Description] Prints data in buffer and feeds the paper in **n** vertical dots. after printing finished, setup next line beginning as initial position (n vertical dot

pitch = 0.125 mm , below is same)

2 Characters setup commands

2.1 ESC ! *n*

[Name] Select print mode

[Format]	ASCII	ESC	!	<i>n</i>
	Hex	1B	21	<i>n</i>
	Decimal	27	33	<i>n</i>

[Range] $0 \leq n \leq 255$

[Description] when select double height and double height in meanwhile, printer prints quadruple size character, character aligns long baseline.

Selects print mode(s) by using *n* as follows:

Bit	value	Function
0	0	Font A ASCII character (12 × 24)
	1	Font B ASCII character (8 × 16)
1,2	-	Undefined
3	0	Turn off bold mode
	1	Turn on bold mode
4	0	Turn off double-height mode
	1	Turn on double-height mode
5	0	Turn off double-width mode
	1	Turn on double-width mode
6	-	Undefined
7	0	Turn off underline mode
	1	Turn on underline mode

2.2 FS ! *n*

[Name] Select Chinese print mode

[Format] ASCII FS ! *n*

Hex 1C 21 *n*

Decimal 28 33 *n*

[Range] $0 \leq n \leq 255$

[Description] when select double height and double height in meanwhile, printer prints quadruple size character, character aligns long baseline.

Selects print mode(s) by using *n* as follows:

Bit	value	Function
0	0	Font A ASCII character (12 × 24)
	1	Font B ASCII character (8 × 16)
1	-	Undefined

2	0	Turn off double-width mode
	1	Turn on double-width mode
3	0	Turn off double-height mode
	1	Turn on double-height mod
4	-	Undefined
5	-	Undefined
6	-	Undefined
7	0	Turn off underline mode
	1	Turn on underline mode

2.3 GS ! n

[Name] Setup character size

[Format] ASCII GS ! n
Hex 1D 21 n
Decimal 29 33 n

[Range] $0 \leq n \leq 255$

($1 \leq$ vertical number of times ≤ 2 , $1 \leq$ horizontal number of times ≤ 8)

[Description] Selects the character height using bits 0 to 3 and selects the character width using

bits 4 to 7, as follows:

bit0	bit1	bit2	bit3	Height
0	0	0	0	1(normal)
1	0	0	0	2 (double width)
0	1	0	0	3 times
1	1	0	0	4 times
0	0	1	0	5times
1	0	1	0	6times
0	1	1	0	7times
1	1	1	0	8times

bit4	bit5	bit6	bit7	Width
0	0	0	0	1(normal)
1	0	0	0	2 (double width)
0	1	0	0	3times
0	0	1	0	4times

0	0	0	0	5times
1	0	1	0	6times
1	1	0	0	7times
1	1	1	0	8times

2.4 ESC – *n*

[Name] turn underline mode on/off

[Format] ASCII ESC - *n*
 Hex 1B 2D *n*
 Decimal 27 45 *n*

[Range] $0 \leq n \leq 2$

[Description] Turns underline mode on/off, based on the following values of *n*:

n (decimal)	Function
0	Turn off underline mode
1	Turn on underline mode(1 dot thick)
2	Turn on underline mode(2 dots thick)

2.5 ESC E *n*

[Name] Turn bold mode on/off

[Format] ASCII ESC E *n*
 Hex 1B 45 *n*
 Decimal 27 69 *n*

[Range] $0 \leq n \leq 1$

[Description] Turn bold mode on/off

n	Function
0	Turn off bold mode
1	Turn on bold mode

2.6 GS B *n*

[Name] Turn white/black reverse printing mode

[Format] ASCII GS B *n*
 Hex 1D 42 *n*
 Decimal 29 66 *n*

[Range] $0 \leq n \leq 1$

[Description] Turn white/black reverse printing mode

n	Function
0	Turn off white/black reverse mode
1	Turn on white/black reverse mode

2.7 ESC V n

[Name] Turn character rotation mode on/off

[Format]

ASCII	ESC	V	n
Hex	1B	56	n
Decimal	27	86	n

[Range] $0 \leq n \leq 1$

n is used as follows:

n	Function
0	Turn off 90° clockwise rotation mode
1	Turn on 90° clockwise rotation mode

[Description] Turn character rotation mode on/off

When printer setup underline mode, 90° clockwise rotation characters won't be underline.

2.8 FS &

[Name] setup Chinese character mode

[Format]

ASCII	FS	&
Hex	1C	26
Decimal	28	38

[Description] setup Chinese character mode

[Notes] this command is only valid when printer supported GB18030 code system.

2.9 FS .

[Name] setup ASCII character mode

[Format]

ASCII	FS	.
Hex	1C	2E
Decimal	28	46

[Description] setup ASCII character mode

[Notes] this command is only valid when printer supported GB18030 code system.

2.10 ESC R n

[Name] Select ASCII character set

[Format]

ASCII	ESC	R	n
Hex	1B	52	n
Decimal	27	82	n

[Range] $0 \leq n \leq 13$

[Description] Selects ASCII character set n from the following table:.

n	Character set
0	U.S.A

1	France
2	Germany
3	U.K
4	Denmark I
5	Sweden
6	Italy
7	Spain I
8	Japan
9	Norway
10	Denmark II

2.11 ESC t n

[Name] Select character codepage

[Format] ASCII ESC t n

Hex 1B 74 n

Decimal 27 116 n

[Description] Selects page n from the character code table.

n	Page
0	PC437[USA-Europe standard]
1	Katakana
2	PC850[multiple language]
3	PC860 [portuguese]
4	PC863 [canadian-french]
5	PC865 [north Europe]
16	WPC1252
17	PC866[slavic2]
18	PC852[latin2]
19	PC858 [Europe]

3 Printing typesetting commands

3.1 ESC 3 n

[Name] Set line spacing

[Format] ASCII ESC 3 n

Hex 1B 33 n

Decimal 27 51 n

[Range] $0 \leq n \leq 255$

[Description] Set the line spacing as n vertical dot pitch (1mm = 8 dots)

3.2 ESC SP n

[Name] Set character right-side spacing

[Format] ASCII ESC SP n

Hex 1B 20 n

Decimal 27 32 n

[Range] $0 \leq n \leq 255$

[Description] Sets the character spacing for the right side of the character to $[n \times 0.125\text{mm}(n \times 0.0049\text{ inch})]$.

[Note] ▪ the right-side character spacing for double-width mode is twice than normal value, When character are enlarged, the characters spacing is n times the normal value

3.3 ESC a n

[Name] Select alignment

[Format] ASCII ESC a n
 Hex 1B 61 n
 Decimal 27 97 n

[Range] $0 \leq n \leq 2$

[Description] The command is enabled only when processed at the beginning of the line in standard mode. This command executes alignment in the printing area, n selects the alignment, as follows::

n	alignment
0	Left alignment
1	Central alignment
2	Right alignment
8	Same line left alignment
10	Same line right alignment

3.4 GS L nL nH

[Name] Setup left margin

[Format] ASCII GS L nL nH
 Hex 1D 4C nL nH
 Decimal 29 76 nL nH

[Range] $0 \leq nL \leq 255$

$0 \leq nH \leq 255$

[Description] left margin is setup as “n” horizontal dot pitch. nL nH are unsigined double byte integral low byte and high byte, $N = nL + nH \times 256$, left margin as the width of available printing area to left side distance.

4 Barcode printing commands

4.1 GS h n

[Name] Set barcode height

[Format] ASCII GS h n
 Hex 1D 68 n
 Decimal 29 104 n

[Range] $1 \leq n \leq 240$ unit:0.125mm

[Description] Sets bar code height
n is vertical dots number

[Notes] If $n > 240$, barcode height is setup as 40

4.2 GS w n

[Name] Set barcode width

[Format]	ASCII	GS	w	n
	Hex	1D	77	n
	Decimal	29	119	n

[Range] $1 \leq n \leq 4$

[Description] Sets horizontal size of the barcode

n definition as follow:

N	Module width (mm) for multi-level barcode	Binary-level barcode	
		Thin element width (mm)	Thick element width (mm)
1	0.125	0.125	0.25
2	0.25	0.25	0.50
3	0.375	0.375	0.75
4	0.50	0.50	1.0

4.3 GS H n

[Name] Select printing position for readable HRI characters

[Format]	ASCII	GS	H	n
	Hex	1D	48	n
	Decimal	29	72	n

[Range] $0 \leq n \leq 2$

[Description] Selects the printing position of readable HRI characters when printing a bar code



n select printing position ,as follow

n	HRI characters Printing position
---	----------------------------------

0	No printing
1	Above the barcode
2	Below the barcode
3	Both above and below bar code

4.4 GS p n

[Name] set barcode alignment

[Format]	ASCII	GS	p	n
	Hex	1D	50	n
	Decimal	29	90	n

[Range] n = 0, 1, 2

[Description] Selects a font for the HRI characters used when printing a bar code.

n selects a font from the following table

n	alignment
0	Left alignment
1	Central alignment
2	Right alignment

4.5 GS k

[Name] Print barcode

[Format] there are two formats:

① ($0 \leq m \leq 8$)

ASCII	GS	k	m	d1...d k	NUL
Hex	1D	6B	m	d1...d k	00
Decimal	29	107	m	d1...d k	0

② ($65 \leq m \leq 73$)

ASCII	GS	k	m	n	d1... dn
Hex	1D	6B	m	n	d1... dn
Decimal	29	107	m	n	d1... dn

[Range] ① $0 \leq m \leq 8$ (k and d depend on the barcode system used)。

② $65 \leq m \leq 73$ (n and d depend on the barcode system used)。

n as printing barcode length

[Description] Selects a barcode system and prints barcode.

m selects a barcode system as follows:

m	Barcode system	Character quantity	remark
0	UPC-A	$11 \leq k \leq 12$	$48 \leq d \leq 57$

①	1	UPC-E	$11 \leq k \leq 12$	$48 \leq d \leq 57$
	2	JAN13 (EAN13)	$12 \leq k \leq 13$	$48 \leq d \leq 57$
	3	JAN8 (EAN8)	$7 \leq k \leq 8$	$48 \leq d \leq 57$
	4	CODE39	$1 \leq k$	$48 \leq d \leq 57, 65 \leq d \leq 90$ 32.36.37.43.45.46.47
	5	ITF	$1 \leq k$ (k as even number)	$48 \leq d \leq 57$
	6	CODABAR	$1 \leq k$	$48 \leq d \leq 57, 65 \leq d \leq 68$ 36.43.45.46.47.58
	7	Standard EAN13	$12 \leq k \leq 13$	$48 \leq d \leq 57$
	8	Standard EAN8	$7 \leq k \leq 8$	$48 \leq d \leq 57$
	65	UPC-A	$11 \leq n \leq 12$	$48 \leq d \leq 57$
	66	UPC-E	$11 \leq n \leq 12$	$48 \leq d \leq 57$
	67	JAN13 (EAN13)	$12 \leq n \leq 13$	$48 \leq d \leq 57$
	68	JAN8 (EAN8)	$7 \leq n \leq 8$	$48 \leq d \leq 57$
	69	CODE39	$1 \leq n \leq 255$	$48 \leq d \leq 57, 65 \leq d \leq 90$ 32.36.37.43.45.46.47
	② 70	ITF	$1 \leq n \leq 255$ (n as even number)	$48 \leq d \leq 57$
	71	CODABAR	$1 \leq n \leq 255$	$48 \leq d \leq 57, 65 \leq d \leq 68$ 36,43,45,46,47,58
	72	CODE93	$1 \leq n \leq 255$	$0 \leq d \leq 127$
	73	CODE128	$1 \leq n \leq 255$	$0 \leq d \leq 127$
	74	Standard EAN13	$12 \leq n \leq 13$	$48 \leq d \leq 57$
	75	Standard EAN8	$7 \leq n \leq 8$	$48 \leq d \leq 57$

[Notes] if horizontal data beyond printing area, then beyond part will be ignored.

Printer does not calculate check code, if barcode needs check code, then user needs to include check code into barcode data. Printer has no responsible for verifying check code, if user calculates error check codes, then it leads to unreadable barcodes.

If HRI characters is setup printing, CODE93 and CODE128 unreadable characters won't be printed out and replaced as "□"

CODE39 doesn't include external code CODE39 (EXTERN CODE 39)

CODE93 doesn't include external code CODE93 (EXTERN CODE 93)

CODE128 the top of the bar code data string must be the code set selection character (CODE A、CODE B and CODE C). Special characters are defined by combining two characters "{" and one character. The ASCII character "{" is defined by transmitting "{" twice consecutively. As follow :

ASCII	HEX	Function
{A	7B, 41	Code A
{B	7B, 42	Code B

{C	7B, 43	Code C
{S	7B, 53	SHIFT
{1	7B, 31	FNC1
{2	7B, 32	FNC2
{3	7B, 33	FNC3
{4	7B, 34	FNC4

5 Bitmap printing commands

5.1 GS v 0 m xL xH yL yH d1...dk

[Name] Print raster bitmap

[Format] ASCII GS v 0 m xL xH yL yH d1...dk

Hex 1D 76 30 m xL xH yL yH d1...dk

Decimal 29 118 48 m xL xH yL yH d1...dk

[Range] $0 \leq m \leq 3$, $1 \leq xL \leq 128$

$xH=0, 1 \leq (xL + xH*256) \leq 128$ $0 \leq yL \leq 255$

$1 \leq yH \leq 151 \leq (yL + yH*256) \leq 4095$ $0 \leq d \leq 255$

$k = (xL + xH*256) * (yL + yH*256)$ ($k \neq 0$)

[Description] set raster bitmap m value as below table:

m	Format	Vertical density	Horizontal density
0	normal	203.2dpi	203.2dpi
1	Double-width	203.2dpi	101.6dpi
2	Double-height	101.6dpi	203.2dpi
3	Four times	101.6dpi	101.6dpi

dpi : dots/25.4mm(1 inch)

[Note] XL, xH sets the bitmap horizontal directional data bytes $(xL + xH*256)$;

YL, yH sets the bitmap vertical data bytes $(yL + yH*256)$

In standard mode, this command is valid only if there is no data in the print buffer.

For raster bitmap print, the command is not affected by the print mode (character size, bold, double-strike, reverse printing, underline print mode) .

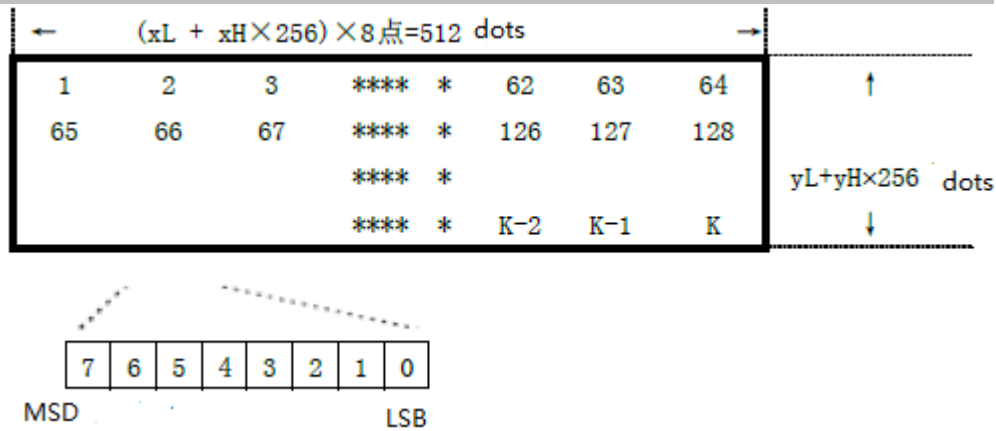
[Detailed description]

- If the width of the print area set by the GS L (setting the left pitch) is less than the minimum print width, the printer extends only the problematic rows to the minimum width. The minimum width to normal mode is $m=0$ and the high mode $m=2$ is one point, to the double width mode $m=1$ and four times the size mode $m=3$ is two points.

- The data within the print area is read in and discarded by point by point. If the print position of the subsequent character is a multiple of 8. The print position of the characters to be printed as raster bitmap, followed by HT (horizontal tabulation), ESC \$(set absolute print position), ESC \ (set relative print position) and GS L (setting left distance). The ESC a (setting alignment) setting is also valid for raster bitmaps. When the command is received during the macro definition, the printer ends the macro definition and executes the command, which should clearly define the command.

- The D specifies the bitmap data, with the print point corresponding to 1 and the print point corresponding to 0.

[Example] When $xL+xH*256=64$



5.2 FS q n [xL xH yL yH d1 ... dk] 1 ... [xL xH yL yH d1 ... dk] n

[Name] define NV bitmap

[Format] ASCII FS q n [xL xH yL yH d1...dk] 1... [xL xH yL yH d1...dk] n

Hex 1C 71 n [xL xH yL yH d1...dk] 1... [xL xH yL yH d1...dk] n

Decimal 28 113 n [xL xH yL yH d1...dk] 1... [xL xH yL yH d1...dk] n

[Range] $1 \leq n \leq 255$

$0 \leq xL \leq 255$

$0 \leq xH \leq 3$ ($1 \leq (xL + xH * 256) \leq 1023$)

$0 \leq yL \leq 255$

$0 \leq yH \leq 1$ (当 $1 \leq (yL + yH * 256) \leq 288$)

$0 \leq d \leq 255$

$k = (xL + xH * 256) * (yL + yH * 256) * 8$

Defined total data=192k byte

- N is for the number of NV bitmap defined
- xL, xH : the horizontal dots $(xL + xH * 256) * 8$ of defined NV bitmap
- yL, yH is the vertical dots $(yL + yH * 256) * 8$ of defined NV bitmap

[Note]

- NV storage may be damaged by executing writing commands frequently.

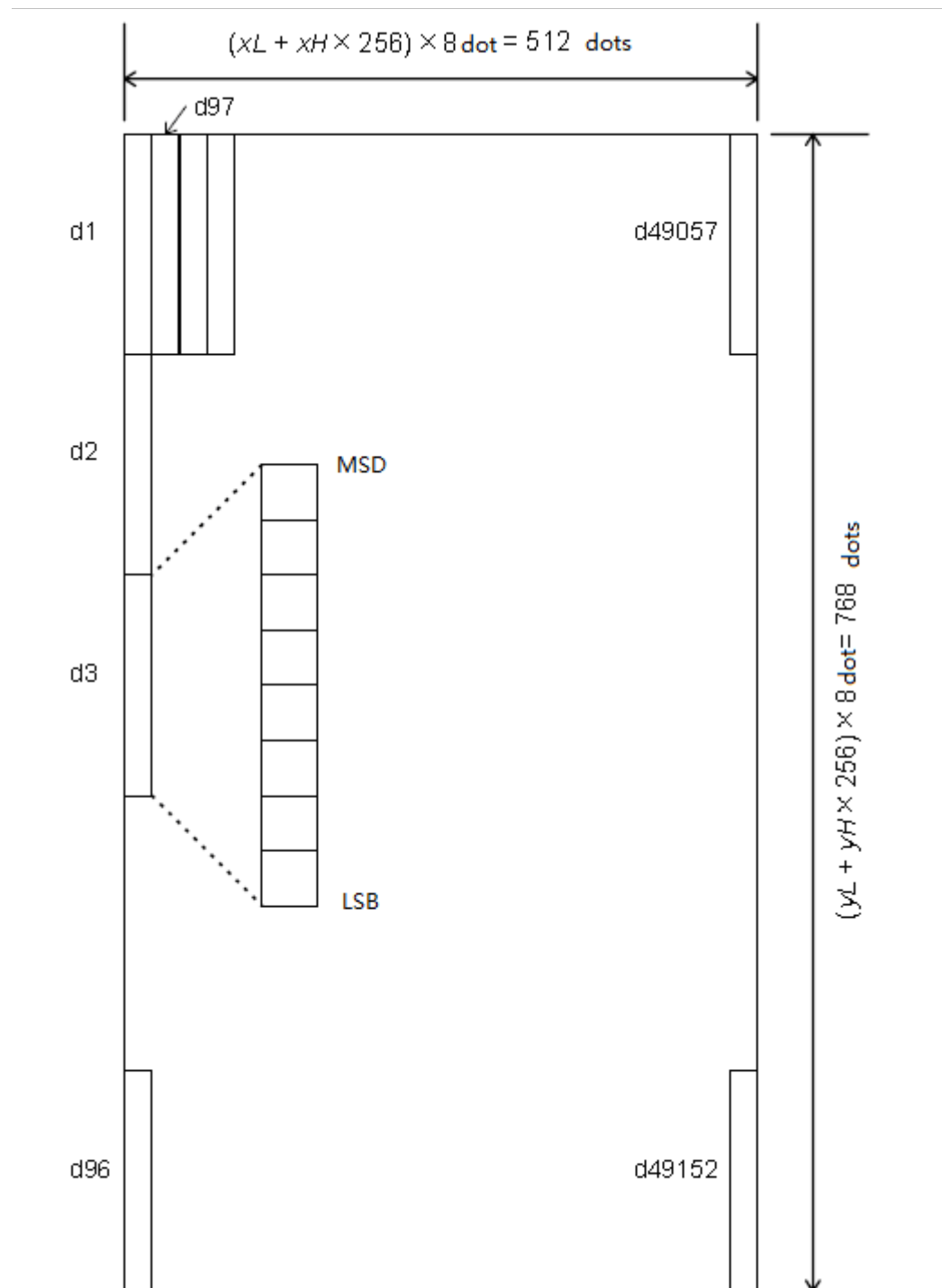
Suggestion is less than 10 times

- After the process of placing an image in the NV memory, the printer performs a hardware reset operation, so the user-defined character, the down bitmap, and the macro should be defined after the completion of the command. The printer clears the receive and print buffers and resets them to the active mode when the power is turned on.
- This command deletes all NV bitmaps that have been defined with this command
- From this command to the completion of the hardware reset during the completion of the hardware can not be carried out (including when the cover is opened to initialize the print head position with paper feed button, etc.)

- During this command processing, the printer is busy and stops receiving data when writing data to the user NV memory. Therefore, it is prohibited to transmit data during the execution of this command, including real-time commands
- In standard mode, the command is only valid at the beginning of a line of processing
- The command's 7 bytes <FS yH> is valid after the normal processing.
- When the amount of data exceeds the left capacity of the range defined by xL, xH, yL, yH, the printer will process the range defined by xL, xH, yL, yH outside the defined range.
- In a set of non-first group of bitmaps, when the printer encounters cases where xL, xH, yL, yH are out of the defined range, the command is stopped and the NV image is written. At this point, the NV bitmap that has not been defined is disabled (undefined, but any previously defined NV bitmap is still valid).
- D in the data (d), a 1-bit specifies a point to be printed and a 0-bit specifies a point that is not printed.
- The command defines n as the number of NV bitmaps. The number starts to rise from bitmap 01H. Thus the first data set [xL xH yL yH d1 ... dk] is the NV bitmap 01H, and the last data set [xL xH yL yH d1 ... dk] is the NV bitmap n. The total number matches the number of NV bitmaps set by the FS p command.
- The definition data of an NV bitmap consists of [xL xH yL yH d1 ... dk]. Thus, when there is only one NV bitmap, n = 1, the printer only processes the data set [xL xH yL yH d1 ... dk] once. The printer uses the NV memory ([data: (xL xH xH -256) - (yL yL yH -256) -8] [header: 4]) bytes.
- The definition area in this printer is 192K bytes (max). The command can define several bitmaps, but can not define bitmaps for total data capacity [bitmap data + head] over 192K bytes.
- Even if the ASB is set, the printer does not transmit ASB status or perform status check during processing of the command Measurement.
- When the command is received during a macro definition, the printer stops the macro definition and starts executing the command.
- Once an NV bitmap is defined, it can not be executed by the ESC @ command, reset, and power is removed.
- This command performs only the definition of the NV bitmap and does not print. The NV bitmap is printed by FS p Implemented.

[Reference]FS p

[Example]xL = 64, xH = 0, yL = 96, yH = 0



5.3 FS p n m

[Name] Print NV bitmap

[Format] ASCII FS p n m

Hex 1C 70 n m

Decimal 28 112 n m

[Range] $1 \leq n \leq 255$

$0 \leq m \leq 3$, $48 \leq m \leq 51$

[Description] Print NV bitmap by m

m	Format	Vertical density	Horizontal density
0, 48	Normal	203.2 dpi	203.2 dpi
1, 49	Double-width	203.2 dpi	101.6 dpi
2, 50	Double-height	101.6 dpi	203.2 dpi
3, 51	4 times	101.6 dpi	101.6 dpi

- n is the number of NV bitmaps (defined with the FS q command).

- m Specifies the bitmap mode.

[Note]

- The NV bitmap is a bitmap defined in nonvolatile memory. Use FS q to define FS p print

- The command is invalid when the specified NV bitmap does not exist.

- In standard mode, the command is valid only if there is no data in the print buffer.

- This command is not affected by the print mode (except for bold, overlay, underline, character size, anti-print, or character 90), rotation, and so on.

If the print area width of the NV bit set by GS L and GS W is less than one vertical line, do the following only for the problem line. In the NV bitmap mode, a vertical line means a point in the normal mode (m? 0, 48) and the multiplier mode (m? 2, 50), the times length mode (m? 1, 49) and the quadruple size mode (M 3, 51) under the two points:

① In NV bitmap mode, the print area width extends to the right by a vertical line, in which case the print area is not exceeded.

② If the print area width can not extend a vertical line, the left margin is reduced to accommodate a vertical line.

- If the down bitmap to be printed exceeds one line, the excess data is not printed.

- In Normal and Multiplier mode, the command feeds n (n is the NV bitmap height), in the double and quad size modes (the command feeds 2n and n is the NV bitmap height) , Regardless of the line spacing set by ESC 2 or ESC 3.

After printing the bitmap, the command sets the print position at the beginning of one line and processes the subsequent data as normal data.

[Reference] ESC *, FS q, GS /, GS v 0

6 printer status feedback commands

6.1 DLE EOT n

[Name] Real-time status transmission

[Format]	ASCII	DLE	EOT	n
	Hex	10	04	n
	Decimal	16	4	n

[Range] n=1, 2, 3, 4,5

[Description] Transmit the selected printer status specified by n in real-time, according to the follow parameters:

- n = 1: Transmit printer status;
- n = 2: Transmit offline status;
- n = 3: Transmit error status;
- n = 4: transmit paper roll sensor status;
- n=5: transmit external sensors status

- [Details]
- Each byte stands for one status when printer transmits its present status
 - Printer doesn't confirm that host can get data when transmitting
 - Printer works after receives this command

• Printer still executes this command under RS-232 port mode working. no matter printer offline, buffer full, or other error status happened.

n = 1 printer status

Bit	Off/On	Hex	Decimal	Function
0	Off	00	0	No used. Fixed to Off
1	On	02	2	No used. Fixed to On
2	On	04	4	No used. Fixed to On
3	Off	00	0	Online
	On	08	8	Offline
4	On	10	16	No used. Fixed to On
5	Off	00	0	Does not wait for online error recovery
	On	20	32	Waits for online error recovery
6	Off	00	0	FEED switch is Off
	On	40	64	FEED switch is On
7	Off	00	0	No used. Fixed to Off

Notice: bit5 is called "online error" is the period that printer executes great commands and self page test command to wait switch

n = 2 offline status

Bit	Off/On	Hex	Decimal	Function
0	Off	00	0	No used. Fixed to Off
1	On	02	2	No used. Fixed to On
2	Off	00	0	Platen is closed
	On	04	4	Platen is opened
3	Off	00	0	Paper is not being fed by FEED button
	On	08	8	Paper is being fed by FEED button
4	On	10	16	No used. Fixed to On
5	Off	00	0	No paper-end stop
	On	20	32	Printing is being stopped
6	Off	00	0	No error
	On	40	64	Error occurred

7	Off	00	0	No used. Fixed to Off
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Notice: bit5 turns on when paper sensor checked paper end and printer stop working

n = 3 error status

Bit	Off/On	Hex	Decimal	Function
0	Off	00	0	No used. Fixed to Off
1	On	02	2	No used. Fixed to On
2	Off	00	0	No mechanical error
	On	04	4	mechanical error has occurred
3	Off	00	0	No auto cutter error
	On	08	8	auto cutter error has occurred
4	On	10	16	No used. Fixed to On
5	Off	00	0	No unrecoverable error
	On	20	32	unrecoverable error has occurred
6	Off	00	0	No auto-recoverable error
	On	40	64	auto-recoverable error has occurred
7	Off	00	0	No used. Fixed to Off

Notice: bit2 when platen be opened and can not find BM also be treated as mechanical error

Bit6 when printer head over heating, bit6 will be opened until the temperature drops down or platen be opened

n = 4 continuous paper sensor status

Bit	Off/On	Hex	Decimal	Function
0	Off	00	0	No used. Fixed to Off
1	On	02	2	No used. Fixed to On
2	Off	00	0	Paper roll end sensor: paper adequate
	On	0C	12	Paper near-end is detected by the paper roll near-end sensor
4	On	10	16	No used. Fixed to On
5	Off	00	0	paper roll sensor: paper present
	On	60	96	paper roll end detected by paper roll sensor
7	Off	00	0	No used. Fixed to Off

n = 5 external sensor status

Bit	Off/On	Hex	Decimal	function
0	off	10	16	External sensor status normal
	on	11	17	Paper blocked
1	off	10	16	External sensor status normal

	on	12	18	Dragging ticket
2	off	14	20	External sensor status
	on	10	16	Ticket out or ticket out sensor is invalid
3~7	-	-	-	unused

6.2 GS r

[Name] Transmit status

[Format] ASCII GS r n

Hex 1D 72 n

Decimal 29 114 n

[Range] n = 1

[Description] Transmits paper sensor status

Paper sensor status n = 1

Bit	Off/On	Hex	Decimal	Function
0	Off	00	0	Paper roll end sensor: paper adequate
1	On	03	3	Paper roll end sensor: paper near end
2	Off	00	0	Paper roll end sensor: paper adequate
3	On	0C	12	Paper roll end sensor: paper end
4	On	10	16	No used. Fixed to On
5 6	-	-	-	Undefined
7	Off	00	0	No used. Fixed to Off

When printer finished last ticket date and send this command, will get a return byte; if there is no return data , then use DLE EOT n to check printer real time status.

7 cutter commands

7.1 ESC i

[Name] full cutting

[Format] ASCII ESC i

Hex 1B 69

Decimal 27 105

[Description] full cutting

7.2 ESC m

[Name] partial cutting

[Format] ASCII ESC m

Hex 1D 6D

Decimal 29 109

[Description] partial cutting

8 page mode commands

8.1 enter page mode

[Name] enter page mode

[Format] ASCII ESC m
 Hex 11 21
 Decimal 17 33

8.2 set page range

[Name] set page range

[Format] ASCII ESC m xH xL yH yL
 Hex 11 23 xH xL yH yL
 Decimal 17 35 xH xL yH yL

[Description]

landscape $0 \leq xH*256 + xL \leq 576$ unit: 0.125mm

portrait $0 \leq yH*256 + yL \leq 640$ unit: 0.125mm

8.3 set print data beginning coordinate

[Name] set print data beginning coordinate, data top alignment

[Format] ASCII ESC m xH xL yH yL
 Hex 11 24 xH xL yH yL
 Decimal 17 36 xH xL yH yL

[Description]

landscape $0 \leq xH*256 + xL \leq 576$ unit: 0.125mm

portrait $0 \leq yH*256 + yL \leq 640$ unit: 0.125mm

8.4 set page printing direction

[Name] set page printing direction

[Format] ASCII ESC m n
 Hex 11 25 n
 Decimal 17 37 n

[Description]

n	direction
0	normal
1	rotation 90°
2	rotation 180°
3	rotation 270°

8.5 print page data

[Name] print complete page data

[Format] ASCII ESC m
 Hex 11 0C
 Decimal 17 12

8.6 exit page mode

[Name] exit page mode

[Format] ASCII ESC m

Hex	11	22
Decimal	17	34